It is recommended that you spend about 30 minutes on this question.

2 A hot wire in air loses energy. Some of the energy is lost to the air particles that hit the wire. This energy lost depends on the number of particles hitting the wire per second, and hence on the air pressure.

If the temperature of the wire is constant then the total energy lost per second is equal to the electrical energy supplied per second.

Design a laboratory experiment to investigate how the total energy lost per second from a wire depends on the air pressure. You should draw a detailed labelled diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) how the electrical energy supplied to the wire per second would be measured,

(c) the type of thermometer used to check that the temperature of the wire remains constant during the experiment,

(d) how the pressure would be changed and measured,

(e) any safety precautions that you would take.

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